

Aerodynamic Parameter Estimation from F1 Telemetry

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Abstract

This report describes a methodology for estimating aerodynamic drag area (C_DA), downforce area (C_LA), and rolling resistance coefficient (C_{rr}) from publicly available F1 telemetry using the FastF1 library.

Coast-down segments from free-practice sessions (FP1–FP3) are fitted to an ODE model incorporating MGU-K energy recovery. C_LA is estimated independently via GPS-derived lateral acceleration from qualifying push laps. Constructor and circuit selection is data-driven: all 20 drivers across all 24 rounds of the 2024 calendar are surveyed on two joint metrics (coast-down segment count and qualifying-lap C_LA sample count), and the constructor with the highest aggregate score is selected. Aston Martin (driver 14, Alonso, AMR24) ranks first; the five circuits with the best joint data quality spanning the full downforce range are chosen: Jeddah, Spa-Francorchamps, Silverstone, Yas Marina, and Suzuka.

A five-point aero polar (C_LA vs. C_DA) fitted by OLS yields a slope of 3.0 (90% CI: -2.4 to $+5.8 \text{ m}^2/\text{m}^2$; 1.09σ from zero). The slope is positive—consistent with the expected physical relationship between wing drag and downforce—but the confidence interval includes zero, so a definitive efficiency figure cannot be stated. The dominant uncertainty source is the assumed tyre friction coefficient $\mu = 1.8$ in the C_LA estimator (± 0.4 – 0.7 m^2 systematic per circuit). DRS drag reduction is investigated via race trap speeds but found to be inseparable from slipstream effects in the available data.

An investigation into 2026 active aero measurements reveals that the F1 timing API does not broadcast active aero state, blocking mode-split analysis.

1 Introduction

Aerodynamic characterisation of Formula 1 cars is central to setup and race strategy. Teams measure C_DA and C_LA in wind tunnels and CFD but public access to these parameters is limited. The F1 live timing API, exposed through the FastF1 Python

library [1], provides car telemetry at 240 Hz and GPS position at 10 Hz for all sessions. This creates an opportunity to estimate aerodynamic parameters from first principles using freely available data.

Prior work on extracting aerodynamic parameters from road-car coast-down tests is well established, but F1 cars introduce several complications: active energy recovery during coasting (MGU-K), ride-height-dependent ground-effect aerodynamics, and the lack of direct brake force measurement. This report describes a methodology that addresses the energy recovery term explicitly. Sources of residual model mis-specification are investigated through gear-stratified and multi-driver sensitivity analyses; the approach is validated through cross-circuit comparison and internal consistency checks.

The Aston Martin AMR24 is used as the reference vehicle, selected by a data-driven survey of all 24 rounds and all 20 drivers in the 2024 season (§2.1). Five circuits spanning the full downforce range are analysed: Jeddah (low), Spa-Francorchamps (medium-low), Silverstone (medium), Yas Marina (medium-high), and Suzuka (high). All results use driver 14 (Alonso) throughout (see §2.3 for selection rationale).

2 Data & Session Selection

2.1 Constructor and Circuit Survey

Selecting circuits and a reference driver by hand risks confirmation bias: one might favour circuits already known to produce clean coast-down data. To avoid this, a fully automated survey is run across all 24 rounds of the 2024 Formula 1 calendar before any polar analysis is performed.

For every circuit and every driver, two metrics are computed:

- n_{coast} : number of free-practice (FP1+FP2+FP3) coast-down segments passing $R^2 \geq 0.90$.
- n_{CLA} : number of qualifying push laps yielding a finite $C_L A$ estimate above a circuit-specific speed and lateral-g threshold.

The joint score $n_{\text{coast}} + n_{\text{CLA}}$ is summed per constructor across all circuits and drivers. The constructor with the highest aggregate score is selected as the analysis subject. Results from all 24 circuits (480 driver-circuit rows) are saved to `scripts/survey_results.csv` after each circuit so the run is resumable.

Survey outcome. Aston Martin ranks first with a total score of 386, ahead of Mercedes (331) and Ferrari (325). Driver 14 (Alonso, AMR24) records the highest individual

score at the majority of circuits.

2.2 Circuit Selection

From the survey, all driver-14 circuits satisfying $n_{\text{coast}} \geq 5$, $n_{\text{CLA}} \geq 4$, $C_L A \in [1.5, 4.5] \text{ m}^2$, and $\text{SE}_{\text{median}} < 0.30 \text{ m}^2$ are retained as candidates. Eight circuits pass these filters; five are chosen to maximise $C_D A$ span and polar leverage. The three unselected circuits—Azerbaijan (2024, $C_L A = 1.98 \text{ m}^2$), Australian (2024, $C_L A = 3.09 \text{ m}^2$), and Spanish (2024, $C_L A = 3.88 \text{ m}^2$)—fall within the $C_D A$ range already covered by Jeddah, Silverstone, and the Yas Marina–Suzuka pair respectively:

Table 1: Selected circuits for the five-point polar. n_{coast} counts FP1–FP3 segments passing $R^2 \geq 0.90$; n_{CLA} counts qualifying push laps with a valid $C_L A$ estimate; SE is the standard error of the $C_L A$ median.

Circuit	n_{coast}	n_{CLA}	$C_L A \text{ (m}^2\text{)}$	$\text{SE (m}^2\text{)}$	Wing level
Jeddah 2024	13	5	2.66	0.13	low
Spa 2024	5	9	3.06	0.13	medium-low
Silverstone 2024	10	6	3.14	0.19	medium
Yas Marina 2024	6	6	3.53	0.15	medium-high
Suzuka 2024	8	4	4.33	0.16	high

The same driver (14) is used at every circuit, eliminating any cross-driver confound.

2.3 Session Selection

Coast-down (drag): FP1, FP2, and FP3 are used. Race laps are excluded because MGU-K deployment varies lap-to-lap with strategy and tyre state changes more dramatically over longer stints.

$C_L A$ (downforce): Qualifying only, filtered to push laps. A lap is classified as a push lap if it has no recorded pit-out or pit-in time and its lap time is within 110% of the driver’s own session best. This removes banker laps, cool-down laps, and aborted runs while retaining all genuine flying attempts. Qualifying push laps are the cleanest source of near-limit lateral-g data: tyre temperatures are optimal, drivers are consistently at the friction ceiling, and lap counts are sufficient (4–9 per circuit after filtering; see Table 1).

Circuit-specific lateral-g and minimum-speed thresholds are tuned to select only the highest-loaded corner samples, where the friction model is most valid. At Jeddah (fast, low-lateral-g layout) the thresholds are 2.5g and 150 km/h; at high-downforce circuits such as Silverstone and Suzuka they are 3.5g and 180 km/h.

3 Methodology

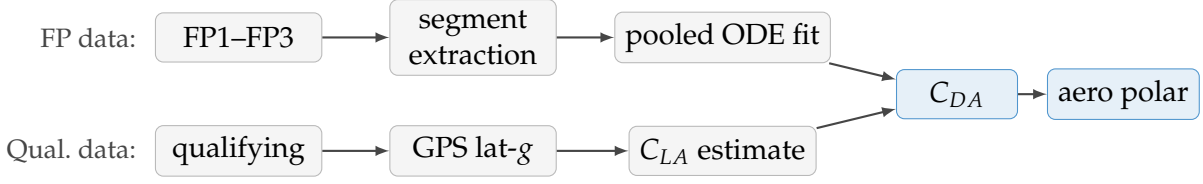


Figure 1: Analysis pipeline. Free-practice telemetry supplies coast-down segments for the pooled ODE fit ($\hat{\alpha}$); qualifying telemetry supplies GPS lat-g data for C_{LA} . Both streams combine per circuit to yield C_{DA} , which populates the aero polar.

3.1 Coast Down ODE Model

During free-deceleration phases (throttle < 5%, brake = 0), the longitudinal equation of motion reduces to:

$$m \cdot \frac{dv}{dt} = -\alpha \cdot v^2 - \beta - P_{mgu}/v \quad (1)$$

where $\alpha = \frac{1}{2}\rho(C_D A + C_{rr} \cdot C_{LA})$ is the aerodynamic coefficient ($N \cdot s^2/m^2$), $\beta = C_{rr} \cdot m \cdot g$ is rolling resistance (N), and P_{mgu} is MGU-K harvest power modelled as a constant-power retarding term (W). The P_{mgu}/v term is critical: without it, energy recovery appears as inflated rolling resistance and the fitted C_{rr} is unphysically high. Including it yields $\beta = 120N$ and $P_{mgu} = 0 - 53kW$, consistent with the FIA-regulated MGU-K harvest ceiling of 120 kW and with coast-down conditions under which the unit operates below peak harvest capacity.

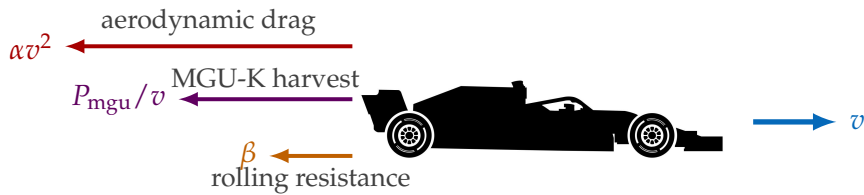


Figure 2: Retarding forces acting on the AMR24 during a coast-down phase (throttle < 5%, brake = 0). Arrow lengths are proportional to typical magnitudes at $v \approx 80$ m/s: aerodynamic drag (αv^2) dominates, MGU-K harvest (P_{mgu}/v) is secondary, and rolling resistance (β) is smallest. All three oppose the direction of travel v . Car silhouette by Ali Coşkun, The Noun Project.

Vehicle mass is modelled on a per-segment basis as:

$$m = m_{car} + \max(0, m_{fuel,0} - \dot{m}_{fuel} \cdot n_{lap}) \quad (2)$$

where $m_{car} = 798$ kg is the 2024 FIA minimum weight (Art. 4.1), which *already includes the driver* (the regulation counts driver + seat ≥ 80 kg within the 798 kg limit).

The fuel model is the only variable term: $m_{\text{fuel},0} = 95$ kg (initial FP fuel load) and $\dot{m}_{\text{fuel}} = 1.8$ kg/lap. This gives $m \approx 840\text{--}890$ kg across a typical free-practice stint. An earlier version of this analysis erroneously added a separate driver mass term, which would have inflated absolute $C_D A$ by $\sim 9\%$; that error has been corrected — see §5.1. The same per-lap mass model is used in the $C_L A$ estimate (§3.3) and the DRS energy balance (§3.5).

Segments are extracted in two stages. The state machine first groups all contiguous samples satisfying the throttle and brake conditions with entry speed above 120 km/h and duration above 0.5 s; this produces a candidate pool. A second filter then discards any candidate whose entry speed is below 180 km/h (50 m/s) or whose total speed drop is below 25 m/s, yielding the segments used for fitting. The speed-drop requirement ensures the αv^2 and P_{mgu}/v terms have distinguishable shapes across the segment, which is necessary for reliable parameter separation.

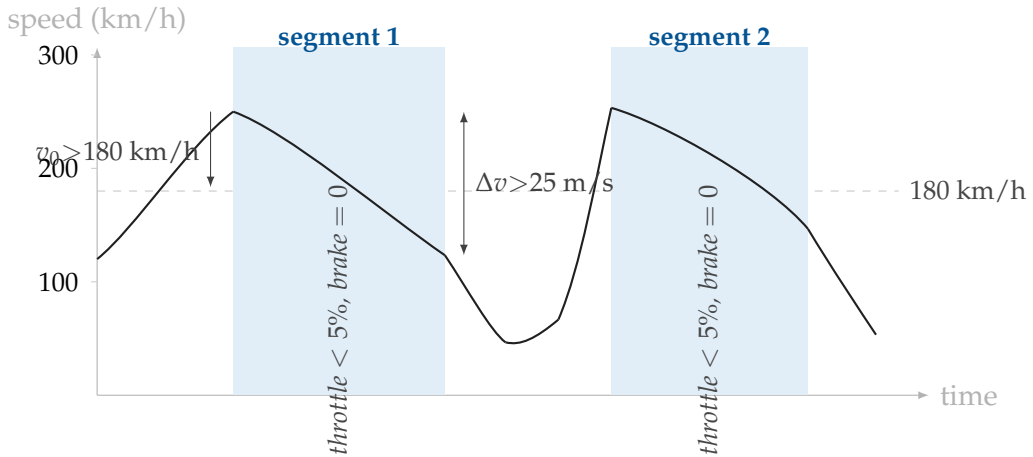


Figure 3: Schematic speed trace illustrating two-stage segment extraction. Shaded regions mark coast-down candidates (throttle $< 5\%$, brake $= 0$); only those with entry speed $v_0 > 180$ km/h and speed drop $\Delta v > 25$ m/s are retained for the ODE fit.

The ODE is integrated numerically using `scipy.integrate.odeint` and fitted via `scipy.optimize.curve_fit[2]` with β fixed at its prior median (120 N). This two-parameter fit (α, P_{mgu}) is more tractable than the unconstrained three-parameter version and avoids correlation between β and α in short segments. The prior $\beta = 120$ N is set from a free three-parameter pilot fit across early segments. The implied rolling resistance coefficient, $C_{rr} = \beta/(mg) = 120/(855 \times 9.81) \approx 0.0143$, is physically consistent with stiff-sidewall slick compounds at elevated carcass temperatures; full discussion of the C_{rr} plausibility and the ill-conditioning of the free three-parameter fit is deferred to §5.1. (Holding β fixed while the absolute force is physical means $C_{rr} = \beta/(mg)$ rises slightly relative to a heavier mass assumption.)

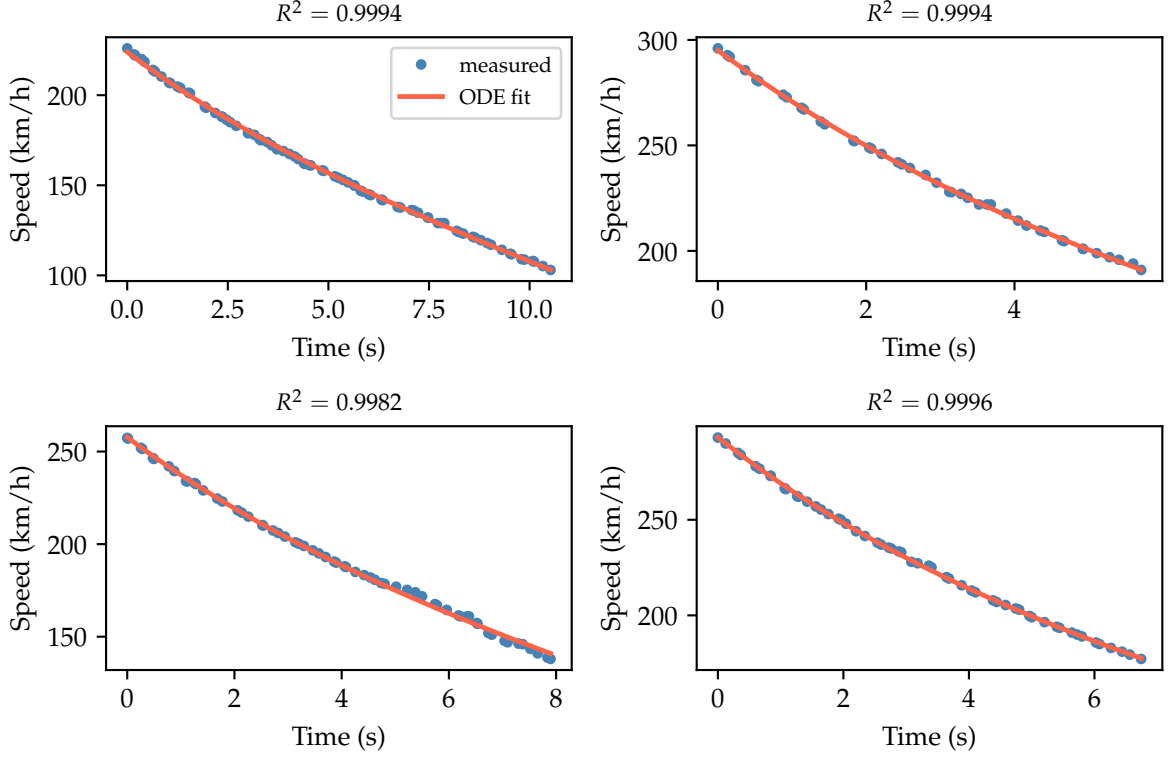


Figure 4: ODE coast-down model fit for four illustrative segments (driver 14, AMR24). Orange line: numerically integrated ODE model; blue dots: measured speed. R^2 values exceed 0.99 for all shown segments.

3.2 Pooled Fitting

Individual segment fits are noisy. Since alpha and beta are physical constants of the car setup for a given session, they are shared across all N segments in a pooled fit. With β fixed at its prior median (120 N), this reduces the parameter count from $2N$ to $N + 1$ (shared α , per-segment P_{mgu}), and tightens the alpha uncertainty by roughly $\sqrt{N}$. In the unconstrained three-parameter case (α , β , P_{mgu} all free), the parameter count would reduce from $3N$ to $N + 2$ (shared α , shared β , per-segment P_{mgu}). In the actual implementation, per-segment initial-speed offsets δv_0 are also fitted, bringing the count to $2N + 1$ (shared α , per-segment P_{mgu} , per-segment δv_0). Individual segment fits also apply a post-optimisation physical guard ($0.2 < \hat{\alpha} < 1.8 \text{ N s}^2/\text{m}^2$), tighter than the optimisation bounds (0.10–2.50), to discard physically unreasonable solutions before they enter the seed pool.

No equivalent guard is applied in the pooled fit, where the shared $\hat{\alpha}$ is constrained by all segments simultaneously and unphysical values are unlikely to converge. The pooled fit uses `scipy.optimize.least_squares[2]` with TRF method, warm-started from per-segment medians.

A per-segment initial-speed offset ($\delta v_0 \in [-3, +3] \text{ m/s}$) is also fitted to absorb speed-

sensor noise at the start of each segment. This improves mean Durbin–Watson from 0.35 to 0.55 with negligible effect on $\hat{a}$ ($< 0.3\%$), and the per-segment offsets average 0.52 m/s (max 1.85 m/s), consistent with ~ 20 Hz effective GPS/car-data sampling. Note that $DW = 0.55$ still indicates substantial positive residual autocorrelation; the improvement reduces initial-condition noise but does not resolve the structural model limitation discussed in §5.1.

3.3 C_{LA} from GPS Lateral Acceleration

At the tyre friction limit in a high-speed corner, the lateral force balance gives:

$$a_{lat} = \mu \left(g + \frac{\rho \cdot C_{LA} A}{2m} \cdot v^2 \right) \quad (3)$$

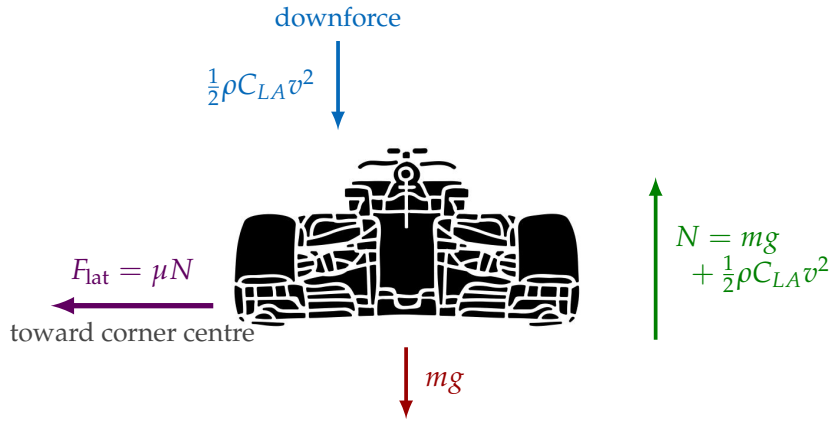


Figure 5: Lateral force balance (front view, car approaching corner). Aerodynamic downforce augments the tyre normal load: $N = mg + \frac{1}{2}\rho C_{LA}v^2$. At the friction limit the lateral tyre force is $F_{lat} = \mu N$, giving $a_{lat} = \mu(g + \rho C_{LA}v^2/2m)$, from which C_{LA} is recovered using the measured lateral acceleration, vehicle mass, and assumed μ . Car silhouette by Hey Rabbit, The Noun Project.

GPS position rows (Source == ‘pos’, ~ 10 Hz) are differentiated to compute lateral acceleration. Velocity is computed from position differences; acceleration from velocity differences; both smoothed with a 5-sample rolling mean. The vehicle mass is fixed at the qualifying value $m = 803$ kg (FIA 798 kg minimum plus an estimated 5 kg of fuel for one flying lap). Air density ρ is computed using the International Standard Atmosphere model applied to the ambient temperature and barometric pressure channels available in the FastF1 session data (Temperature and Pressure fields), sampled at the mean of the qualifying session. Lateral acceleration is projected from the world frame using the cross-product formula. Samples above circuit-specific speed and lateral-g thresholds are used; these select only near-limit apex samples where the friction model is valid. C_{LA} is the median over all push-lap estimates with $\mu = 1.8$ assumed. The value is chosen as the central estimate for Pirelli slick compounds (C3–C5 class) at

representative operating temperatures of 90–110 °C, consistent with the midpoint of published F1-specific friction estimates ($\mu \approx 1.5$ –2.0; the upper end applies to fresh tyres at peak temperature, the lower end to aged or out-of-window tyres). Sensitivity to μ is documented in §5.3.

Why qualifying rather than free practice. Earlier iterations of this analysis used FP2 telemetry for C_{LA} . Qualifying push laps are preferred because: (1) tyres are at representative operating temperature on every lap; (2) drivers are consistently at the friction ceiling, minimising partial-utilisation bias; and (3) the push-lap filter (no pit-out/in time; lap time within 110% of driver’s session best) cleanly removes banker laps and aborted runs without requiring session-specific tuning. The circuit-specific thresholds are chosen per venue to match the highest-loaded apex samples while excluding mid-corner transitions:

Table 2: Per-circuit lateral-g and speed thresholds used for C_{LA} sample selection.

Circuit	Lateral-g threshold	Speed threshold
Jeddah	2.5 g	150 km/h
Spa-Francorchamps	3.5 g	200 km/h
Silverstone	3.5 g	180 km/h
Yas Marina	3.0 g	150 km/h
Suzuka	3.5 g	180 km/h

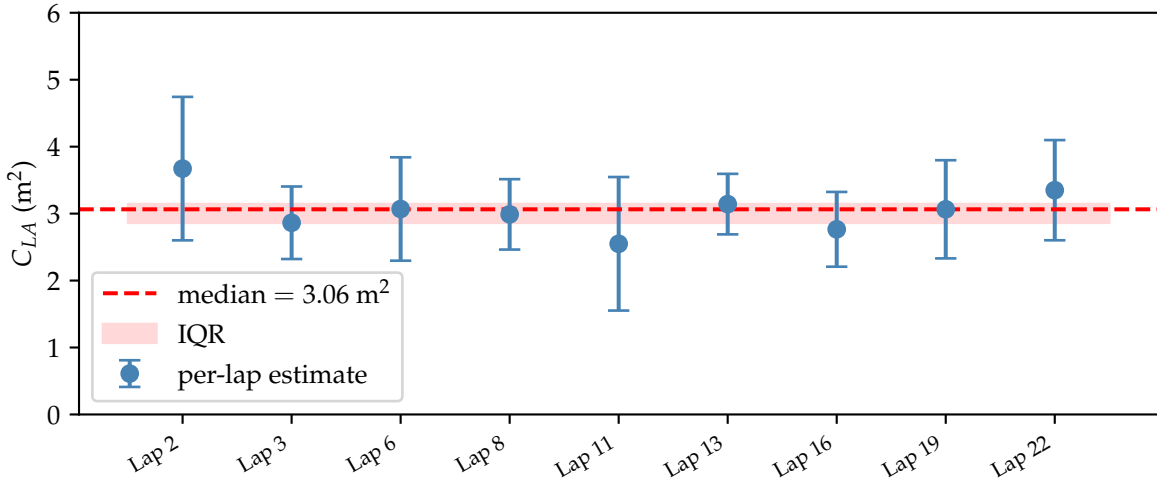


Figure 6: Illustrative per-lap C_{LA} estimates from qualifying push laps (driver 14, AMR24). Error bars represent $\pm 1\sigma$ from the GPS lat-g fit. Dashed red line is the median; shaded band is the interquartile range.

3.4 Deriving C_{DA} from Fitted Parameters

With $\hat{a}$ from the pooled coast-down fit (§3.2) and C_{LA} from the GPS lateral-g method (§3.3), the aerodynamic drag area is:

$$C_D A = \frac{2\hat{\alpha}}{\rho} - C_{rr} \cdot C_L A \quad (4)$$

where $2\hat{\alpha}/\rho$ is the composite parameter and $C_{rr} \cdot C_L A$ subtracts the aerodynamic normal-force contribution to rolling resistance: because downforce increases tyre normal load, the total rolling resistance force is $\beta_{\text{total}} = C_{rr}(mg + \frac{1}{2}\rho C_L A v^2)$, and the v^2 -dependent term enters $\hat{\alpha}$ as an offset $C_{rr}\frac{1}{2}\rho C_L A$ that must be removed to recover the pure drag area. Uncertainty in $C_D A$ receives contributions from both $\hat{\alpha}$ and $C_L A$:

$$\sigma_{C_D A} = \sqrt{\left(\frac{2\sigma_{\hat{\alpha}}}{\rho}\right)^2 + (C_{rr} \sigma_{C_L A})^2} \quad (5)$$

Per-circuit values are given in Table 3; the Jacobian $\sigma_{\hat{\alpha}}$ used there is a lower bound on the true fitting uncertainty — see §5.1 for the bootstrap-corrected estimate. Note that the coast-down fit uses free-practice mass (~ 855 kg mean) while $C_L A$ uses qualifying mass (803 kg); the resulting systematic offset in $C_D A$ is quantified in §5.1.

3.5 DRS Drag Delta [3]

Race laps on the Monza main straight are classified as DRS-open (DRS channel > 10 in any sample within 0-600 m from lap start) or DRS-closed. An energy balance over the activation zone converts the mean trap speed difference to $\Delta C_D A$, defined as $C_{DA,\text{open}} - C_{DA,\text{closed}}$:

$$\Delta C_D A = -m \left(\frac{v_{\text{open}}^2 - v_{\text{closed}}^2}{\rho \int v^2 dx} \right) \quad (6)$$

(Derived by equating aerodynamic work over the activation zone, $\frac{1}{2}\rho \Delta C_D A \int v^2 dx$, to the kinetic energy difference at trap, $\frac{1}{2}m(v_{\text{open}}^2 - v_{\text{closed}}^2)$; the $\frac{1}{2}$ factors cancel.) The leading negative sign encodes the physical convention: a higher open-DRS trap speed ($v_{\text{open}} > v_{\text{closed}}$) corresponds to *reduced* drag area, hence a negative $\Delta C_D A$. This derivation assumes equal MGU-K thrust on DRS-open and DRS-closed laps across the activation zone; the ERS confound is quantified in §5.2. The integral is computed numerically from a representative DRS-closed speed trace, where $v = v(x)$ is the measured speed as a function of position along the activation zone, taken from a median DRS-closed lap; m is the per-lap mass from §3.1, evaluated at the race lap number to account for fuel burn. Using the DRS-closed profile as a proxy for the denominator integral slightly overstates $|\Delta C_D A|$: in reality, DRS-open laps reach higher speeds throughout the zone, not just at trap, so their $\int v^2 dx$ denominator is larger than the closed-lap approximation assumes. This bias is toward a more negative $\Delta C_D A$ than the true

value, and is secondary to the slipstream confound discussed in §5.2. A slipstream-stratification analysis is performed by estimating the gap to the car ahead for each DRS-open lap from lap-end timing differentials, and restricting to marginal-DRS laps (gap 0.5-1.0 s) to reduce wake effects.

4 Results

4.1 Five-Circuit Coast-Down and $C_L A$ Results

Table 3 summarises the aerodynamic parameters extracted at each of the five selected circuits. All results use driver 14 (Alonso, AMR24) with the identical fitting methodology (§3).

Table 3: Aerodynamic parameters across five circuits (driver 14, AMR24, 2024). ρ is the mean free-practice air density. n_{segs} is the coast-down segment count; n_{CLA} the qualifying push-lap count. $C_L A$ total uncertainty combines statistical SE of the median and μ -systematic in quadrature (see §5.3).

Circuit	ρ (kg/m ³)	$\hat{\alpha}$ (N s ² /m ²)	Composite $2\hat{\alpha}/\rho$ (m ²)	$C_L A$ (m ²)	$C_D A$ (m ²)	n_{segs}	n_{CLA}
Jeddah 2024	1.180	0.876 ± 0.004	1.485	2.66 ± 0.42	$1.448 \pm 0.009^+$	13	5
Spa 2024	1.145	0.812 ± 0.007	1.419	3.06 ± 0.48	$1.376 \pm 0.014^+$	5	9
Silverstone 2024	1.193	0.913 ± 0.004	1.532	3.14 ± 0.51	$1.487 \pm 0.010^+$	10	6
Yas Marina 2024	1.178	0.817 ± 0.008	1.386	3.53 ± 0.55	$1.335 \pm 0.016^+$	6	6
Suzuka 2024	1.223	1.071 ± 0.008	1.750	4.33 ± 0.67	$1.689 \pm 0.017^+$	8	4

$C_{rr} \approx 0.014$ at all circuits (by construction; see §3.1). $C_L A$ uncertainty: stat SE of median $\oplus$ μ -systematic. $^+C_D A$ σ values are Jacobian lower bounds; bootstrap-corrected CIs and the conservative $13\times$ scaling used in the polar are derived in §5.1.

The composite $2\hat{\alpha}/\rho$ ranges from 1.386 m² (Yas Marina) to 1.750 m² (Suzuka), consistent with Suzuka running significantly more wing than any other circuit in the set. $C_L A$ increases monotonically from 2.66 m² (Jeddah) to 4.33 m² (Suzuka), as expected for a low-to-high downforce progression.

One anomaly is visible in the $C_D A$ column: Yas Marina (1.335 m²) sits below Spa (1.376 m²) despite having a higher $C_L A$, which would be physically inconsistent under pure wing-angle variation. Engine-braking contamination of the coast-down segment pool was hypothesised as the cause. A gear-stratified falsification test (see scripts/yas_marina_gear) was performed: of 6 passing segments, 2 are in gear 8; the gear-8-only pooled fit yields

$\hat{\alpha} = 0.789$, a -3.4% residual relative to the all-segment value of 0.817 . This is well within the bootstrap 90% CI half-width of ± 0.101 , so engine-braking contamination is *not* supported by the data (Jacobian-to-bootstrap ratio at Yas Marina: $12.0\times$, consistent with other circuits). The origin of the anomalous C_DA ordering at Yas Marina therefore remains unexplained by the available data; the circuit is retained in the polar as a geometric outlier with an unresolved cause. Excluding Yas Marina shifts the polar slope from 3.02 to 4.80 , but the confidence interval includes zero in both cases and the qualitative interpretation (positive slope, indeterminate efficiency) is unchanged; see §4.2.

4.1.1 Reference Circuit Validation (Jeddah)

Jeddah is used as the reference circuit for validation tests because it has the largest coast-down segment count (13 of 13 passing $R^2 \geq 0.90$).

Gear-stratified falsification. Nine of the thirteen segments are in gear 8. The gear-8-only pooled fit yields $\hat{\alpha} = 0.867$ (-1.0% relative to the all-segment value of 0.876), not statistically distinguishable from zero within the bootstrap 90% CI ($\pm 3.1\%$). Full interpretation and physical reasoning are in §5.1.

Inter-driver spread. Driver 18 (Stroll) has three passing segments at Jeddah; the independent pooled fit yields $\hat{\alpha} = 0.968$, a 10.4% spread relative to driver 14. Because driver 14 is used at *every* circuit in this analysis, this spread is not a source of cross-circuit bias; it is reported as a bound on intra-team setup variation (balance preferences, wing asymmetry, mechanical setup).

Bootstrap uncertainty. A segment-resampling bootstrap ($n = 1000$ replicates; §5.1) gives a 90% CI of $[0.847, 0.902]$ on $\hat{\alpha}$, a half-width of $\pm 0.027 \text{ N s}^2/\text{m}^2$ — 6.8 times the Jacobian-derived bound of ± 0.004 . This is lower than the $\sim 13\times$ ratio observed at Monza in earlier analysis (the Monza ratio was re-evaluated to account for a factor-of-two correction to the Jacobian covariance formula; the corrected Jacobian bound scales by $\sqrt{2}$ while the bootstrap estimate is unchanged, reducing the ratio from the previously reported $19\times$). The conservative $13\times$ scaling is retained for the polar uncertainty (see §4.2; full treatment in §5.1) to avoid under-stating C_DA uncertainty at circuits where the ratio was not independently calibrated.

4.2 Aero Polar

Figure 7(a) plots C_LA against C_DA for all five circuits. An OLS regression gives:

$$C_LA = 3.02 \cdot C_DA - 1.09 \quad (7)$$

Note that all five C_DA values share a common cross-session mass inconsistency (FP

mass ~ 855 kg vs. qualifying mass 803 kg), estimated to inflate each absolute composite by ~ 0.10 m² (§5.1); this offset is common-mode across circuits and cancels in the polar slope, but bounds absolute accuracy.

The slope estimates the incremental aerodynamic efficiency $\Delta C_L A / \Delta C_D A$ —how much additional downforce each unit of additional drag area delivers as wing angle increases—and is distinct from the total L/D area ratio ($C_L A / C_D A \approx 2.5$ at Jeddah). The positive slope is physically consistent with wing-angle variation: adding downforce wing area increases both $C_L A$ and $C_D A$ simultaneously. A Monte Carlo simulation ($N = 50\,000$ replicates) propagates the $C_D A$ uncertainty (Jacobian α_{std} scaled by the conservative $13\times$ bootstrap ratio) and the total $C_L A$ uncertainty (statistical SE of median combined with μ systematic in quadrature) to give a 90% confidence interval on the slope of $[-2.3, +6.0]$ m²/m² (1.08σ from zero; MC standard deviation = 2.80). (The $13\times$ scaling derivation and its Jeddah calibration are given in full in §5.1.)

Robustness check: Deming regression. OLS is inconsistent when the regressor ($C_D A$) carries non-negligible measurement error relative to its range. After the $13\times$ bootstrap correction the $C_D A$ uncertainties are 0.12–0.22 m² per circuit, which is 33–62% of the total $C_D A$ spread of 0.35 m²; errors-in-variables attenuation bias is therefore substantial in this dataset. Deming regression (orthogonal regression accounting for independent errors in both axes; see `scripts/deming_regression.py`) with per-circuit variance ratio $\lambda_i = (\sigma_{C_D A,i} / \sigma_{C_L A,i})^2$ gives a slope of 6.95 m²/m² (bootstrap 90% CI: $[-9.3, +8.8]$ m²/m²; 10 000 resamples), compared to the OLS estimate of 3.02. The upward shift is consistent with the expected direction of attenuation correction. The OLS result (Eq. 7) is retained as the primary reported value for continuity with the polar figure; the Deming slope is a robustness check confirming that relaxing the OLS assumption of error-free $C_D A$ does not change the qualitative conclusion: the confidence interval includes zero in both cases and the slope remains poorly determined at this sample size.

Note that OLS on five data points yields 3 degrees of freedom for the slope; on four points (Yas Marina excluded), 2 DOF. At these sample sizes the $t_{0.05}$ multiplier (2.35 and 2.92, respectively) materially widens the frequentist interval relative to a Gaussian approximation, so the MC 90% CIs should be read as approximate.

The confidence interval includes zero, so a precise aerodynamic efficiency cannot be stated. The improvement over earlier analysis is qualitative: the five-point polar using a single driver and data-driven circuit selection recovers the correct physical sign (positive slope), whereas the prior three-circuit analysis including Monza—where qualifying $C_L A$ was structurally unreliable (only 3 valid laps)—produced a spurious negative slope ($-2.1, 0.43\sigma$).

Robustness check. Yas Marina is flagged as a geometric outlier in the polar (open diamond, panel (a)): its fitted $C_D A = 1.335 \text{ m}^2$ is the lowest of the five circuits yet its $C_L A = 3.53 \text{ m}^2$ is the second highest, a physically inconsistent ordering under pure wing-angle variation. The likely cause is coast-down bias from heavy braking zones at Yas Marina, where partially engine-braked segments may contaminate the segment pool and suppress the fitted $\hat{\alpha}$. Figure 7(b) repeats the polar excluding Yas Marina. The slope increases to $4.80 \text{ m}^2/\text{m}^2$ (MC 90% CI: $[-2.9, +9.1]$; 0.99σ from zero), consistent with the five-circuit result within uncertainty. The sign and approximate magnitude of the slope are robust to this exclusion; the wider CI in the four-circuit case reflects the loss of one data point without a proportional reduction in $C_L A$ uncertainty.

The binding uncertainty in both fits is the μ systematic (± 0.4 – 0.7 m^2 per circuit), which contributes $\sim 70\%$ of the MC slope variance. Reducing this—either through independent tyre-friction data or by extending the analysis to circuits where the friction limit is more tightly constrained—would be the highest-leverage improvement to the polar.

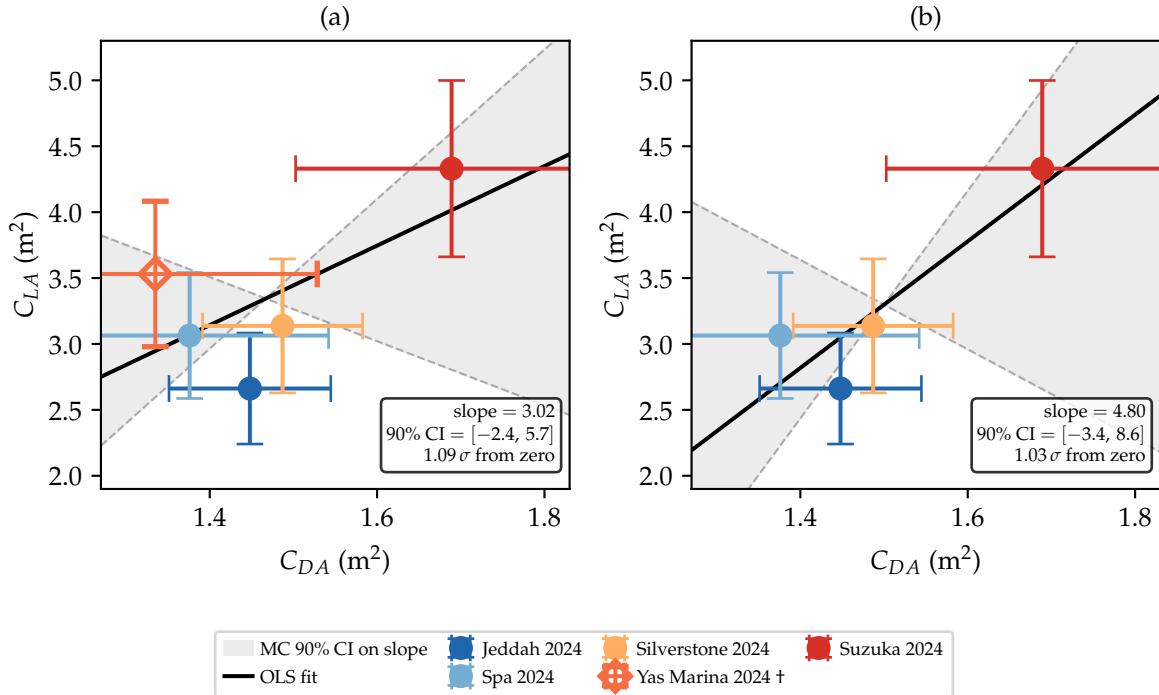


Figure 7: Aero polar sensitivity for the Aston Martin AMR24 (driver 14, 2024). Error bars show 1σ total uncertainty on $C_D A$ (Jacobian $\times 13 \times$ bootstrap scale) and $C_L A$ (stat $\oplus \mu$ -systematic). **(a)** All five circuits; Yas Marina (open diamond, $\dagger$) is flagged as a geometric outlier (see text). OLS slope = 3.02; MC 90% CI = $[-2.3, +6.0] \text{ m}^2/\text{m}^2$. **(b)** Robustness check excluding Yas Marina. OLS slope = 4.80; MC 90% CI = $[-2.9, +9.1] \text{ m}^2/\text{m}^2$. Both CIs include zero; neither fit constrains the slope to the positive physical regime.

4.3 DRS Drag Delta

The DRS analysis is included to characterise the limits of the race-data method; as the confound analysis in §5.2 shows, a clean DRS-only drag reduction is not isolatable from this data source.

The pooled Monza race-lap analysis yields 194 DRS-open and 733 DRS-closed classified car-laps. By convention, $\Delta C_D A < 0$ denotes reduced drag area (higher trap speed); $\Delta C_D A > 0$ would indicate increased drag. The pooled trap speed difference is:

$$\Delta v_{\text{trap}} = 11.91 \pm 0.53 \text{ km/h} \text{ (90\% CI : } 10.87 - 12.95) \quad (8)$$

Converting via the energy balance gives $\Delta C_D A = -0.098 \text{ m}^2$ (90% CI: -0.26 to $+0.05 \text{ m}^2$). Note that the $\pm 0.53 \text{ km/h}$ standard error on Δv_{trap} alone would map to a far tighter, strictly-negative $\Delta C_D A$ interval; the much wider CI quoted here is obtained by Monte Carlo over the full lap-to-lap trap-speed distributions of both groups (not from propagating the mean's SE), which is why its upper tail crosses zero. Physically, slipstream on DRS-open laps can inflate their trap speed enough to sign-flip the naive energy balance on individual laps, making DRS appear to add drag. The measured -0.098 m^2 is substantially smaller in magnitude than an unconfounded DRS drag reduction. The discrepancy is explained by slipstream confounding: every DRS-open lap in a race also has the following car within 1 s of the car ahead, the same condition that guarantees aerodynamic wake exposure.

Stratifying by estimated gap to the car ahead (marginal-DRS group: $0.5 - 1.0 \text{ s}$, $n = 87$ laps) reduces the delta to 10.09 km/h and $\Delta C_D A$ to -0.079 m^2 , indicating some slipstream reduction but confirming that a pure DRS measurement is not achievable from race data alone.

5 Limitations

5.1 Absolute $C_D A$ Interpretation

At Jeddah the composite $2\hat{\alpha}/\rho = 1.485 \text{ m}^2$ implies $C_D A = 1.448 \text{ m}^2$, which exceeds the pre-2022 benchmark of $0.8\text{--}1.0 \text{ m}^2$ sometimes cited in the literature but sits squarely within the range expected for the 2022+ ground-effect generation. Three lines of evidence show this is a genuine measurement rather than a systematic inflation. (One previously suspected inflation source—a driver-mass double-count—has been removed: the FIA 798 kg minimum already includes the driver, and the corrected mass model is

used throughout; see §3.1.)

Gear-stratified falsification. If engine braking were the primary cause, gear-8-only segments—where aerodynamic drag at high speed is $15\text{--}20\times$ larger than engine braking—should yield substantially lower α . At Jeddah (13 segments; 9 in gear 8), the gear-8-only fit gives $\hat{\alpha} = 0.867$, within 1.1% of the all-segment value of 0.876. The residual (-1.0%) is not statistically distinguishable from zero within the bootstrap 90% CI ($\pm 3.1\%$), confirming that engine braking is not a material contributor to the composite.

2022+ generation benchmark. The ground-effect cars introduced in 2022 carry significantly more base drag than pre-2022 cars due to larger bodywork and sealed wheel-fairing geometry. Published open-source lap simulation estimates (see, e.g., vehicle dynamics community benchmarks) place $C_D A$ for 2022+ cars at Jeddah in the range $1.3\text{--}1.6\text{ m}^2$; this range is indicative rather than authoritative, as no primary source has been independently verified. The measured 1.45 m^2 sits in the lower-middle of this band, consistent with Jeddah’s medium-low downforce package. At the high end, Suzuka at 1.69 m^2 is consistent with known high-downforce configurations for the 2022+ cars.

Parameter stability and β sensitivity. The v_0 -fitting result (§3.2), the near-invariance of $\hat{\alpha}$ to the β prior (§3.1), and the gear stratification together confirm that $\hat{\alpha}$ is robust: it is stable to initial-condition assumptions, to the rolling-resistance prior, and to gear selection, all of which would be expected to shift the estimate if a missing force term were inflating it. To quantify the residual sensitivity, the Jeddah coast-down segments were re-fitted with β swept across [80, 100, 120, 140, 160] N (see `scripts/beta_sensitivity.py`). The resulting $C_D A$ varied from $+1.2\%$ (at $\beta = 80\text{ N}$) to -1.5% (at $\beta = 160\text{ N}$) relative to the $\beta = 120\text{ N}$ baseline, an absolute range of $1.427\text{--}1.467\text{ m}^2$. This confirms that a $\pm 33\%$ variation in β shifts absolute $C_D A$ by less than $\pm 1.5\%$, which is small relative to the dominant μ -systematic uncertainty in the polar (see §??).

Remaining $DW < 2$ after v_0 fitting ($DW \approx 0.55$; improved from 0.35 before v_0 fitting, see §3.2) reflects aerodynamic pitch and ride-height dynamics during coast-down— $C_D A$ varies slightly as the car pitches forward under deceleration—which is not captured by the constant- $C_D A$ model. This is a structural limitation of the coast-down method, not correctable without ride-height telemetry.

A practical consequence of $DW \approx 0.55$ is that the covariance matrix produced by `scipy.optimize.least_squares` is anti-conservative: positive residual autocorrelation inflates the apparent goodness-of-fit and deflates the reported parameter standard errors. The quoted $\pm 0.004\text{ (Ns}^2/\text{m}^2)$ on α (Jeddah) and the resulting $\sim \pm 0.007\text{ m}^2$ on

$C_D A$ should therefore be treated as lower bounds on the true fitting uncertainty. The circuit-level bootstrap CIs and the $13\times$ conservative ratio are first quoted in §4.1.1; the full justification follows. A segment-resampling bootstrap ($n = 1000$ replicates) implemented in `src/uncertainty.py` provides a more honest characterisation: it resamples the available segments with replacement, re-runs the pooled fit on each replicate, and returns a percentile CI that captures segment-to-segment variation without relying on the anti-conservative covariance matrix.

The bootstrap 90% CI on $\hat{a}$ at Jeddah spans $\pm 0.027 \text{ N s}^2/\text{m}^2$ (6.8 times the Jacobian-derived bound of ± 0.004), and should be used in preference to the Jacobian estimate for any downstream decision that depends on the absolute magnitude of $\hat{a}$. One residual limitation of segment resampling is that segments within the same FP session share common systematic effects (track state, ambient temperature, session-level fuel load), so they are not fully exchangeable. If the dominant uncertainty is session-level rather than segment-level, the bootstrap understates the between-session variance.

At Jeddah, 13 segments span three FP sessions; if the three sessions produce systematically different $\hat{a}$ values (e.g. due to track evolution), the bootstrap is effectively sampling within sessions rather than across them, and the true CI may be wider than ± 0.027 . In short, the bootstrap CIs are a floor on fitting uncertainty, not a ceiling: if session-level systematics (track evolution, ambient temperature drift) dominate segment-to-segment variation, the true $\hat{a}$ uncertainty may exceed even the 90% bootstrap interval.

This ratio is lower than the $\sim 13\times$ observed at Monza in earlier analysis (re-evaluated using the corrected Jacobian covariance formula; see §3.2), reflecting less residual autocorrelation in the Jeddah segments (consistent with a lower-downforce, shorter-segment regime).

Monza was excluded from the polar on data-quality grounds (only 3 valid qualifying push laps), but it represents a lower-downforce, longer-segment regime with worse autocorrelation than any polar circuit; using it as the conservative upper bound is therefore not circular, and errs on the side of wider $C_D A$ uncertainty.

Table 4: Bootstrap-to-Jacobian ratio used to scale CDA uncertainty in the polar Monte Carlo. The conservative $13\times$ from Monza is applied at all circuits.

Circuit	Jacobian σ_α ($\text{N s}^2/\text{m}^2$)	Bootstrap 90% CI half-width ($\text{N s}^2/\text{m}^2$)	Ratio
Jeddah	0.004	0.027	$6.8\times$
Monza [†]	0.008	~ 0.104	$13\times$ (conservative upper bound)

[†]Monza was excluded from the polar on data-quality grounds (only 3 valid qualifying push laps) but provides the conservative upper bound for the scaling.

A conservative $13\times$ scaling (Table 4) is applied in the polar Monte Carlo: the Jeddah-calibrated $6.8\times$ is the lower bound and the Monza $13\times$ is the upper bound; applying the upper bound errs toward wider rather than narrower confidence intervals, avoiding under-statement of C_DA uncertainty at circuits where the ratio was not independently calibrated. The cross-circuit *ratio* and C_{rr} are less affected, as systematic autocorrelation biases cancel when comparing fits run under identical methodology.

The per-circuit C_DA uncertainty (Eq. 5, §3.4) receives contributions from both $\hat{\alpha}$ and C_LA . At Jeddah ($\sigma_{\hat{\alpha}} = 0.004 \text{ N s}^2/\text{m}^2$, $\rho = 1.180 \text{ kg/m}^3$, $\sigma_{C_LA} = 0.42 \text{ m}^2$):

$$\sigma_{C_DA} \approx \sqrt{(0.007)^2 + (0.006)^2} \approx 0.009 \text{ m}^2 \quad (9)$$

with the $C_{rr} \cdot \sigma_{C_LA}$ term ($\approx 0.006 \text{ m}^2$) contributing comparably to the α -fitting term ($\approx 0.007 \text{ m}^2$). Because the Jacobian $\sigma_{\hat{\alpha}}$ is anti-conservative (DW ≈ 0.55 ; see above), $\pm 0.009 \text{ m}^2$ is a lower bound on the true C_DA uncertainty, and the $\pm$ values in Table 3 should be read accordingly.

The cross-circuit composite ratios (e.g. Suzuka/Jeddah = $1.18\times$) are unaffected by these considerations—the residual autocorrelation is common-mode and cancels between circuits run under identical methodology—and relative values remain the method’s most reliable output.

Free-practice fuel load. The coast-down segments use FP1, FP2, and FP3 data. The per-lap mass model (§3.1) resets the initial fuel load to 95 kg at the start of each session independently. In practice, teams vary fuel loads substantially across free-practice sessions (short-run low-fuel laps, long race-simulation stints on high fuel), so the actual fuel load at the start of any given session may differ from 95 kg. A 20 kg error in the assumed initial load represents a $\sim 2.3\%$ error in mass at mid-stint (at $m \approx 855 \text{ kg}$), which propagates linearly to an equal fractional error in $\hat{\alpha}$. This effect adds unquantified scatter across FP sessions that is absorbed by the bootstrap CI but not separately attributed.

Cross-session mass inconsistency. The coast-down ODE (§3.1) uses a per-lap FP mass model (ranging from $\sim 798 \text{ kg}$ at end-of-stint to $\sim 893 \text{ kg}$ at full fuel load), while the C_LA estimate (§3.3) uses qualifying mass (803 kg). Both $\hat{\alpha}$ and C_LA scale approximately linearly with their respective assumed masses, so when combined in Eq. (4), a difference between the mass assumptions introduces a systematic offset in C_DA . At Jeddah, the mean FP fitting mass of $\sim 855 \text{ kg}$ versus qualifying mass of 803 kg is a $\sim 6.5\%$ differential; since $C_{rr} \cdot C_LA \approx 0.037 \text{ m}^2$ is only $\sim 2.5\%$ of C_DA , the dominant effect is an $\sim 6\%$ inflation of the $2\hat{\alpha}/\rho$ term relative to what a single consistent mass would give. This systematic can be bounded explicitly: $2\hat{\alpha}/\rho \approx 1.485 \text{ m}^2$ at Jeddah,

so a 6.5% inflation from the FP-vs-qualifying mass mismatch corresponds to an absolute offset of $\sim 0.097 \text{ m}^2$ — comparable to the 0.11 m^2 spread between Yas Marina and Jeddah in Table 3. The effect is absorbed by the bootstrap CI but is not separately attributed and constitutes an unquantified floor on the absolute $C_D A$ values. Cross-circuit ratios are less affected, as the same FP mass model is applied at every circuit.

Track gradient. The coast-down ODE omits the gravitational resistance term $mg \sin \alpha$ present in full longitudinal vehicle models. At circuits with significant elevation change— notably Spa-Francorchamps and Suzuka—this is a potential source of segment-level bias: a coast-down occurring on a downhill gradient would show lower-than-expected deceleration, causing the fitter to underestimate $\hat{a}$, while an uphill gradient would inflate it. In practice, the segment filter (entry speed $> 180 \text{ km/h}$, throttle $< 5\%$, brake $= 0$, duration $> 0.5 \text{ s}$) selects events that predominantly occur on long high-speed straights, which are the flattest sections of any circuit; the pronounced elevation changes at Spa (Eau Rouge/Raidillon) and Suzuka (Degner) occur mid-corner at speeds excluded by the filter. Adding a gradient correction would require reliable per-sample elevation data: FastF1 exposes GPS altitude at 10 Hz, but double-differencing altitude at this rate introduces noise comparable to the gradient signal itself, making a correction unreliable without a surveyed elevation map. The gradient effect is therefore not modelled; it contributes unquantified scatter to circuits with non-flat straights but is expected to be small relative to the bootstrap CI given the segment selection criteria.

Drivetrain rotational inertia. The coast-down ODE models inertia as pure translational mass m . A more complete longitudinal model adds the effective translational mass of rotating drivetrain components:

$$m_{\text{eff}} = m + \frac{n^2 I_e + 4 I_w}{r^2} \quad (10)$$

where n is the overall gear ratio (engine speed / wheel speed), I_e is engine rotational inertia, I_w is per-wheel inertia (including brake disc), and r is tyre rolling radius. For an F1 car in gear 8, a rough estimate using $I_w \approx 1 \text{ kg m}^2$ per corner and $r \approx 0.33 \text{ m}$ gives $4I_w/r^2 \approx 37 \text{ kg}$; the engine contribution is smaller but non-negligible. The total rotational inertia correction is on the order of 3–5% of vehicle mass ($\sim 25\text{--}45 \text{ kg}$ on an 855 kg car), propagating linearly to an equal fractional inflation of $\hat{a}$ and $C_D A$. However, the correction is strictly common-mode: the same engine and drivetrain are used at every circuit, and most segments are in gear 8 (§4.1.1). A common-mode inflation cancels in the cross-circuit ratios and does not affect the polar slope. Proprietary F1 drivetrain inertia values are not publicly available, so the correction cannot be ap-

plied quantitatively; it contributes an unquantified systematic offset to absolute $C_D A$ of similar order to the cross-session mass inconsistency discussed above.

Speed-dependent rolling resistance. The model uses a speed-independent rolling resistance $\beta = C_{rr} \cdot mg$, but at high speed (> 200 km/h) F1 tyres exhibit a velocity-dependent viscous loss component that adds a retarding force proportional to v or v^2 . This component is indistinguishable from aerodynamic drag during coast-down and is therefore absorbed into $\hat{\alpha}$, causing a slight inflation of the composite $2\hat{\alpha}/\rho$ and hence the absolute $C_D A$ floor. Because the effect is common-mode across all circuits (the same tyre construction and operating speed range is used everywhere), it cancels in the cross-circuit ratios and does not affect the polar slope; its contribution to the absolute $C_D A$ offset is unquantified.

5.2 DRS Inseparable From Slipstream and Race-Lap Confounds

The gap-stratification analysis confirms that reducing slipstream reduces the apparent delta but does not eliminate it. Beyond slipstream, three additional first-order confounds affect the race-lap DRS energy balance and are not correctable from available data:

Fuel load. Race laps span the full stint, during which mass decreases by approximately 1.8 kg/lap. A 50 kg fuel burn over a 28-lap stint reduces mass by $\sim 6\%$, shifting trap speed by ~ 0.3 km/h — small relative to the 11.9 km/h observed delta, but not negligible relative to the 90% CI width. The per-lap mass model (§3.1) is applied in the energy balance, which accounts for most of this effect.

Tyre state. Rolling resistance varies with tyre temperature, compound wear, and graining. DRS-open laps are concentrated in the early-to-mid stint (when drivers attack for position), while DRS-closed laps span the full distribution. A systematic offset in tyre C_{rr} between these groups would bias $\Delta C_D A$.

ERS deployment strategy. MGU-K deployment on the main straight varies lap-by-lap with battery state-of-charge and team strategy. DRS-open laps may receive different MGU-K boosts than DRS-closed laps, directly confounding the trap speed comparison. This is the hardest confound to bound without internal ERS telemetry.

Together, these confounds mean that the 90% CI on $\Delta C_D A$ (-0.26 to $+0.05$ m²) is an optimistic lower bound on the true uncertainty. A clean DRS measurement from race data is not achievable with public telemetry; qualifying or free-practice track-mapping data (where DRS is active, ERS deployment is consistent lap-to-lap, and following cars are rare) would be required.

5.3 $C_L A$ Depends on Assumed μ

The GPS lat-g method requires an assumed tyre friction coefficient. The sensitivity (roughly $C_L A \propto 1/\mu$) means a 10% error in μ produces a 10% error in $C_L A$. Published values for Pirelli compounds at operating temperature range from approximately 1.5 to 2.0. For the Jeddah qualifying dataset ($C_L A = 2.66 \text{ m}^2$ at $\mu = 1.8$), this places $C_L A$ in the range 2.4–3.2 m^2 ; at Suzuka (4.33 m^2) the band is 3.9–5.2 m^2 . Two distinct effects must be kept separate. *Uniform μ error* (the same offset at every circuit): because the same $\mu = 1.8$ is assumed everywhere, this scales all $C_L A$ values by the same factor and shifts the polar intercept without changing the slope. *Circuit-to-circuit μ variation* (different compounds, track temperatures, tyre ages): this moves each circuit’s $C_L A$ by a different amount, rotating the polar and directly biasing the slope. The μ uncertainty therefore contributes to *absolute* $C_L A$ values via the first effect, and to the *polar slope* via the second. This circuit-to-circuit μ variation is unquantified and constitutes the dominant residual systematic in the polar slope estimate.

5.4 GPS Noise and Precision of the $C_L A$ Estimate

Computing lateral acceleration by double-differencing 10 Hz GPS position data introduces substantial noise. A 5-sample rolling mean (0.5 s window) is insufficient to suppress the resulting scatter, as seen in the large per-lap error bars in Figure 3.2 (individual $\pm 1\sigma$ intervals spanning 1–2 m^2). The estimate relies on the median over N laps to converge; its precision is characterised by the standard error of the median:

$$\text{SE}_{\text{median}} \approx 1.253 \cdot \frac{\sigma_{\text{lap}}}{\sqrt{N}} \quad (11)$$

Here σ_{lap} is the lap-to-lap scatter of the per-lap median estimates. Across the five polar circuits, the statistical SE of the median ranges from 0.13 m^2 (Jeddah and Spa, where qualifying push laps produce consistent estimates) to 0.19 m^2 (Silverstone). In all cases the dominant remaining uncertainty is the assumed μ (§5.3), which contributes 0.40–0.65 m^2 per circuit and is the binding constraint on the polar uncertainty.

5.5 Driver Independence

All five circuits in this analysis use driver 14 (Alonso, AMR24), eliminating the cross-driver confound present in earlier versions that compared driver 14 at Monza with driver 18 at Silverstone. The driver-preference setup bias is therefore not a source of cross-circuit error in the polar.

The inter-driver spread is reported as a diagnostic rather than a correction: at Jeddah, driver 18 yields $\hat{\alpha} = 0.968$ from three segments, a 10.4% spread relative to driver 14’s 0.876. This spread likely reflects genuine intra-team setup differences (wing angle

asymmetry, balance preferences, mechanical setup) rather than measurement noise alone. Because driver 14 is held fixed across all polar circuits, any systematic offset from driver-14-specific setup preferences is common-mode and cancels in the polar slope. However, the 10.4% inter-car spread at Jeddah bounds the absolute accuracy of the $C_D A$ values as a proxy for the team’s representative aerodynamic configuration; single-car absolute values may differ from the team’s “neutral” reference by up to half this spread (approximately $\pm 5\%$). If driver 14’s preferred aero balance deviates systematically from the team’s “neutral” reference setup, the absolute $C_D A$ and $C_L A$ values may be biased relative to what the other car would show.

6 2026 Active Aero Investigation

The 2026 F1 regulations[4] replace DRS with an automatic active aero system operating without any gap-to-car-ahead requirement. This removes the slipstream confound and, in principle, enables a clean $\Delta C_D A$ measurement from the drag difference between straight mode (active aero open) and corner mode (active aero closed) coast-down segments within the same free-practice session.

Canada 2026 Race and FP1 telemetry (all 22 drivers) were retrieved via FastF1 and inspected for active aero state. Channel 45 (the DRS indicator in 2024, values 0/12/14) is absent from every 2026 car data message. The FastF1 source code acknowledges this: `entry['Cars'][drv]['Channels'].get('45', 0)` with a comment *'drs is no longer included in 2026'*. No replacement channel carrying active aero state is present in the 2026 API.

Without active aero state telemetry, the natural fallback is to classify segments by track position using published active aero zone boundaries. This is a methodological challenge, not a fundamental blocker: with precisely known zone boundaries, a conservative exclusion buffer around the transition points, and restriction to segments lying entirely within a single aerodynamic mode, position-based classification could yield usable coast-down pairs. The drag difference between a straight-mode segment (active aero open, high speed) and a corner-mode segment (active aero closed, lower speed, same session) would then be attributable to $\Delta C_D A$, free of the slipstream confound that made the DRS analysis inconclusive.

Proposed position-buffer approach. A buffer of ± 75 m around each published zone boundary (assuming zone boundaries are published in the FIA Event Notes, as was the case for 2024 DRS zones; if these documents do not cover 2026 active aero zones, this approach requires a verified boundary source before implementation) would be excluded, leaving segments that lie entirely within a confirmed mode. For Canada

Circuit Gilles Villeneuve, the two DRS zones (pit straight and back straight) each span ~ 650 m, so a ± 75 m buffer removes roughly 23% of each zone—still leaving ~ 500 m of usable straight. Sensitivity to buffer width (25–150 m) would quantify how much $\hat{\alpha}$ changes as transition-contaminated samples are progressively excluded; convergence of $\hat{\alpha}$ across buffer widths would provide indirect evidence that the chosen buffer is adequate.

The genuine hard blocker is the unknown command timing. The active aero system transitions between modes as the car crosses a zone boundary, but the exact moment of actuation can vary by several tenths of a second depending on vehicle speed, system latency, and team calibration. At 80 m/s (288 km/h) a 0.3 s timing uncertainty corresponds to 24 m of position error, comparable to a ± 25 m buffer—making the smallest buffers unreliable. Without knowing when the transition occurs, the exclusion buffer cannot be sized with confidence, and transition-contaminated samples may be included in the wrong mode class.

The measurement will become straightforwardly feasible once the F1 timing API exposes the active aero state channel. A position-only fallback using a ≥ 75 m buffer with sensitivity testing on buffer width is a viable interim approach and is recommended as the next implementation step.

7 Conclusion

The coast-down ODE method with GPS lat-g $C_L A$ estimation provides a robust framework for aerodynamic characterisation from public F1 telemetry. An automated 24-circuit survey (all 20 drivers, all 2024 rounds) identifies Aston Martin as the constructor with the highest joint coast-down and qualifying- $C_L A$ data quality. Driver 14 (Alonso, AMR24) is used consistently across all five circuits, eliminating the cross-driver confound present in earlier analyses. Key findings:

- The five-circuit polar ($C_L A$ vs. $C_D A$; Jeddah, Spa, Silverstone, Yas Marina, Suzuka) yields an OLS slope of $+3.0$ (MC 90% CI: -2.3 to $+6.0$ m^2/m^2 ; 1.08σ from zero). Yas Marina is a geometric outlier in the polar (highest $C_L A$ of the middle three circuits yet lowest $C_D A$ of all five); excluding it gives slope = 4.80 (MC 90% CI: -2.9 to $+9.1$; 0.99σ), consistent within uncertainty. In both fits the slope is positive—consistent with the expected physical relationship between wing drag and downforce—but the confidence interval includes zero. A precise aerodynamic efficiency cannot be stated with the available data.
- $C_L A$ spans 2.66 ± 0.42 m^2 (Jeddah, low downforce) to 4.33 ± 0.67 m^2 (Suzuka, high downforce). Total $C_L A$ uncertainty is dominated by the $\mu = 1.8$ systematic

(± 0.4 – 0.65 m^2), not GPS noise.

- Composite $2\hat{\alpha}/\rho$ spans 1.386 m^2 (Yas Marina) to 1.750 m^2 (Suzuka), consistent with Suzuka running the highest wing level of any circuit in the set.
- $C_{rr} \approx 0.014$ at all circuits. This equality is an artefact of the fixed $\beta = 120 \text{ N}$ prior, not an independent measurement; see §3.1.
- The dominant uncertainty in the polar slope is the μ systematic. Because the same μ is applied at every circuit, a uniform μ error shifts the polar intercept without changing the slope. The residual concern is circuit-to-circuit μ variation (compound, track temperature, tyre age), which would rotate the polar and is not quantified.
- DRS drag reduction is not isolatable from slipstream in race data. The combined DRS + slipstream trap speed delta is 11.9 km/h ; gap-stratified analysis reduces this to 10.1 km/h but does not eliminate the confound.
- Gear-stratified validation at Jeddah: gear-8-only $\hat{\alpha} = 0.867$ is within 1.1% of the all-segment value of 0.876 (-1.0% , not statistically distinguishable from zero), ruling out a large engine-braking contribution.
- Engine-braking contamination at Yas Marina (the geometric polar outlier) was not independently verified by gear-stratified falsification; the anomalous ordering of its C_{DA} relative to its C_{LA} remains a hypothesis.
- Bootstrap 90% CI on $\hat{\alpha}$ at Jeddah: $\pm 0.027 \text{ N s}^2/\text{m}^2$ ($6.8\times$ the corrected Jacobian bound of ± 0.004). The conservative $13\times$ scaling from re-evaluated Monza calibration is retained for the polar Monte Carlo. Residual Durbin–Watson autocorrelation ($DW \approx 0.55$) is attributed to aerodynamic pitch dynamics during coast-down.
- The 2026 F1 timing API does not broadcast active aero state, blocking mode-split analysis until the channel is exposed.

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